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January 28, 2000

RECEIVED

JAN 28 2000

Christina Kamnikar
Colorado Division of Minerals and Geology
1313 Sherman St., Room 215
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Division of Minerals & Geology

Subject: Response to Technical Adequacy Review of Technical Revision 001,
Lyons Quarry, Permit No. M-77-208, Southdown, Inc.

Dear Christina:

This letter and attachments constitute Southdowns' response to your letter of December 17, 1999.

Item No. 1 – Previous Submittals

The previous submittals by Southdown are hereby included into this Technical Revision.
✓ These submittals are a report on storage of cement kiln dust dated June 21, 1999 and a follow up to that report dated September 29, 1999.

Item No. 2 – Monitoring Program

The monitoring point and the compliance point for ground water monitoring shall be the same point and that shall be water in A-Pit pond. Figure 1 (attached) shows the location of the sampling points. The location is the northwestern shoreline of the pond. This was selected rather than the discharge point to the east because it is closer to the dust disposal area and provides an early warning should a contaminant release be occurring. The water in A-Pit pond is close to and down gradient of the dust disposal area, is on Southdowns' property, and will provide the earliest indication of potential off-site migration of contaminants to the alluvial or bedrock aquifers in the area.

Attached to this letter as Appendix 1, are the water quality analysis methods and quality control/quality assurance methods to be used in the sampling. The parameters to be monitored are as follows:

- Thallium
- Selenium
- Field and lab pH
- Field and lab conductivity

The sampling shall occur twice annually and a report of findings shall be submitted to the DMG with the annual report.

In regard to Harry Posey's memorandum of December 17, 1999, his item number 2 requests that the operator commit to spraying the freshly deposited CKD with water to promote crystallization and inhibit leaching. Southdown has been voluntarily doing this for about two years (along with mixing the CKD with water in a pug mill at the cement plant prior to transport to the disposal area) and will hereby commit to these practices in this Technical Revision. The pug mill mixing occurs all year while the spraying at the disposal area occurs during times of year when the temperature is generally above freezing.

Item No. 3 – Installation of Interim Compacted Shale Cover

The applicant proposes to utilize a clayey shale material for the CKD cap material. The material exists as overburden on an adjacent mining site. The material will be excavated, loaded into mining haul trucks, and hauled to a grizzly-hopper for unloading. The grizzly-hopper will transfer the shale onto an overland conveyor. The material will be conveyed and temporarily stockpiled near the reclamation site. The material will be loaded into haul trucks for final transportation. The material dumped by the haul trucks will be spread with a bulldozer in loose lifts not to exceed one-foot thick, prior to compaction. Total thickness of the cap shall not be less than one-foot thick after compaction. Moisture levels will be adjusted by water additions during the conveying process and after placement with a water truck, as necessary. The material will be compacted in-place by wheel-rolling with the large mining equipment. The applicant will utilize additional equipment if adequate compaction cannot be achieved with on-site equipment. The compacted cap will be a minimum of one-foot thick, with surface grading between 1 and 4%.

The material properties, reasonable moisture control and compactive effort outlined above will achieve an adequate level of imperviousness to minimize precipitation from leaching through the CKD into the groundwater. After the final cover is placed topsoil will be applied and the area will be planted as currently permitted and bonded.

Please let me or John Lohr know if you have any questions.

Sincerely,



Paul Banks

PB/kj

CC: John Lohr

Attachments

APPENDIX 1

Water Quality Analysis Methods and Procedures

1. Monitoring Program

1.1 Monitoring and Compliance Points

The monitoring and compliance points are located at the A Pit, a previously mined pit located along the eastern boundary of the plant. The monitoring and compliance points are shown on Figure 1.

1.2 Monitoring Parameters

Water samples collected from either the monitoring or compliance points will be analyzed for the parameters listed in Table 1. These parameters include field and laboratory determination of pH and specific conductance and laboratory determination of selenium and thallium. Table 1 lists the monitoring parameters, the field or laboratory analytical method, method detection limit, reporting units, sample container type, preservation method (as required), and the maximum holding time. The field and laboratory analytical methods are included as Attachment A.

1.3 Water Sampling

Water samples will be collected at the A Pit monitoring and compliance points using the surface water sampling procedure included as Attachment B. The water samples will be collected using either pre-cleaned disposable sampling equipment or sampling equipment that is decontaminated prior to sampling each location. The samples will be placed in pre-cleaned sample bottles supplied by the laboratory that include the appropriate preservative (as required). At a minimum, the sample bottles will be labeled with the sample number, sample date and time, and sampler name at each sample location. After collection, the samples will be placed in a cooler with blue ice and maintained at approximately 4° Centigrade (C) until delivery to the analytical laboratory. The samples will be maintained under chain-of-custody until delivered to the analytical laboratory. The chain-of-custody will establish a written record of sample handling between the sample site and laboratory.

1.4 Field Quality Assurance/Quality Control

Field analyses will be conducted using U. S. Environmental Protection Agency (EPA) approved analytical methods specified in either EPA (1983)¹. Routine field quality assurance (QA)/quality control (QC) will be used for the analyses. Standard solutions traceable to NIST reference standards will be used to calibrate the pH and specific conductivity instruments. A three-point calibration will be used to calibrate the pH instrument. The specific conductivity instrument will be calibrated using a 1,000 $\mu\text{S}/\text{cm}$ specific conductance standard both before and after analyzing the A Pit samples. A record of both the instrument calibration and sample result measurements will be maintained in the field sampling record.

¹ U. S. Environmental Protection Agency, 1983, Methods for Chemical Analysis of Water and Wastes, EPA 600/4-79-020, Revised March 1983.

1.5 *Laboratory Quality Assurance/Quality Control*

All of the laboratory analyses will be conducted using EPA-approved analytical methods specified in either EPA (1983) or EPA (1986). Routine laboratory QA/QC procedures will be used for the analyses. The QC measures will include laboratory blanks, duplicate analyses of A Pit samples, and laboratory control standards. laboratory blanks are used to determine potential laboratory cross-contamination. The duplicate analyses are used to determine the reproducibility of the results. Laboratory control standards are used to determine the accuracy of the results. Matrix spike analyses are used to determine the sample matrix characteristics and percent recovery.

Table 1

Monitoring Parameters, Analytical Methods, Detection Limits, Preservation, and Maximum Holding Times

Parameter	Field or Laboratory Analysis	Analytical Method	Method Detection Limit	Container	Preservation	Maximum Holding Time
pH	Field	EPA 150.1 ²	0.01 standard units	P, G ³	None	24 hours
Specific Conductance	Field	EPA 120.1 ⁴	1% or 1 $\mu\text{S}/\text{cm}^5$	P, G	None	24 hours
pH	Laboratory	EPA 150.1	0.01 standard units	P, G	None	24 hours
Specific Conductance	Laboratory	EPA 120.1	1% or 1 $\mu\text{S}/\text{cm}$	P, G	None	24 hours
Selenium, total	Laboratory	EPA 7740 ⁶	0.005 mg/L	P, G	HNO ₃ to pH < 2	6 months
Thallium, total	Laboratory	EPA 7841 ⁷	0.002 mg/L	P, G	HNO ₃ to pH < 2	6 months

² U. S. Environmental Protection Agency, 1983, pH Method 150.1 (Electrometric) *in* Methods for Chemical Analysis of Water and Wastes, EPA 600/4-79-020, Revised March 1983.

³ P = polyethylene; G = glass.

⁴ U. S. Environmental Protection Agency, 1983, Conductance Method 120.1 (Specific Conductance) *in* Methods for Chemical Analysis of Water and Wastes, EPA 600/4-79-020. Revised March 1983.

⁵ The method detection limit is 1% or 1 $\mu\text{S}/\text{cm}$, whichever is greater.

⁶ U. S. Environmental Protection Agency, 1986, Method 7740, Selenium (Atomic Absorption, Furnace Technique) *in* Test Methods for Evaluating Solid Waste: Physical/Chemical Methods, 3rd Edition, EPA 530/SW-846, November.

⁷ U. S. Environmental Protection Agency, 1986, Method 7841, Thallium (Atomic Absorption, Furnace Technique) *in* Test Methods for Evaluating Solid Waste: Physical/Chemical Methods, 3rd Edition, EPA 530/SW-846, November.

Attachment A

Field and Laboratory Analytical Methods

CONDUCTANCE

Method 120.1 (Specific Conductance, μmhos at 25°C)

STORET NO. 00095

1. Scope and Application
 - 1.1 This method is applicable to drinking, surface, and saline waters, domestic and industrial wastes.
2. Summary of Method
 - 2.1 The specific conductance of a sample is measured by use of a self-contained conductivity meter, Wheatstone bridge-type, or equivalent.
 - 2.2 Samples are preferably analyzed at 25°C. If not, temperature corrections are made and results reported at 25°C.
3. Comments
 - 3.1 Instrument must be standardized with KCl solution before daily use.
 - 3.2 Conductivity cell must be kept clean.
 - 3.3 Field measurements with comparable instruments are reliable.
4. Precision and Accuracy
 - 4.1 Forty-one analysts in 17 laboratories analyzed six synthetic water samples containing increments of inorganic salts, with the following results:

<u>Increment as Specific Conductance</u>	<u>Precision as Standard Deviation</u>	<u>Bias, %</u>	<u>Accuracy as Bias, $\mu\text{mhos/cm}$</u>
100	7.55	-2.02	-2.0
106	8.14	-0.76	-0.8
808	66.1	-3.63	-29.3
848	79.6	-4.54	-38.5
1640	106	-5.36	-87.9
1710	119	-5.08	-86.9

(FWPCA Method Study I, Mineral and Physical Analyses.)

- 4.2 In a single laboratory (EMSL) using surface water samples with an average conductivity of 536 $\mu\text{mhos/cm}$ at 25°C, the standard deviation was ± 6 .
5. References
 - 5.1 The procedure to be used for this determination is found in:
Annual Book of ASTM Standards, Part 31, "Water", Standard D1125-64, p 120 (1976).
Standard Methods for the Examination of Water and Wastewater, 14th Edition, p 71,
Method 205, (1975).

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pH

Method 150.1 (Electrometric)

STORET NO.

Determined on site 00400

Laboratory 00403

1. **Scope and Application**
 - 1.1 This method is applicable to drinking, surface, and saline waters, domestic and industrial wastes.
2. **Summary of Method**
 - 2.1 The pH of a sample is determined electrometrically using either a glass electrode in combination with a reference potential or a combination electrode.
3. **Sample Handling and Preservation**
 - 3.1 Samples should be analyzed as soon as possible preferably in the field at the time of sampling.
 - 3.2 High-purity waters and waters not at equilibrium with the atmosphere are subject to changes when exposed to the atmosphere, therefore the sample containers should be filled completely and kept sealed prior to analysis.
4. **Interferences**
 - 4.1 The glass electrode, in general, is not subject to solution interferences from color, turbidity, colloidal matter, oxidants, reductants or high salinity.
 - 4.2 Sodium error at pH levels greater than 10 can be reduced or eliminated by using a "low sodium error" electrode.
 - 4.3 Coatings of oily material or particulate matter can impair electrode response. These coatings can usually be removed by gentle wiping or detergent washing, followed by distilled water rinsing. An additional treatment with hydrochloric acid (1 + 9) may be necessary to remove any remaining film.
 - 4.4 Temperature effects on the electrometric measurement of pH arise from two sources. The first is caused by the change in electrode output at various temperatures. This interference can be controlled with instruments having temperature compensation or by calibrating the electrode-instrument system at the temperature of the samples. The second source is the change of pH inherent in the sample at various temperatures. This error is sample dependent and cannot be controlled, it should therefore be noted by reporting both the pH and temperature at the time of analysis.
5. **Apparatus**
 - 5.1 pH Meter-laboratory or field model. A wide variety of instruments are commercially available with various specifications and optional equipment.

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- 5.2 Glass electrode.
- 5.3 Reference electrode—a calomel, silver-silver chloride or other reference electrode of constant potential may be used.
NOTE 1: Combination electrodes incorporating both measuring and reference functions are convenient to use and are available with solid, gel type filling materials that require minimal maintenance.
- 5.4 Magnetic stirrer and Teflon-coated stirring bar.
- 5.5 Thermometer or temperature sensor for automatic compensation.
- 6. Reagents
 - 6.1 Primary standard buffer salts are available from the National Bureau of Standards and should be used in situations where extreme accuracy is necessary.
 - 6.1.1 Preparation of reference solutions from these salts require some special precautions and handling⁽¹⁾ such as low conductivity dilution water, drying ovens, and carbon dioxide free purge gas. These solutions should be replaced at least once each month.
 - 6.2 Secondary standard buffers may be prepared from NBS salts or purchased as a solution from commercial vendors. Use of these commercially available solutions, that have been validated by comparison to NBS standards, are recommended for routine use.
- 7. Calibration
 - 7.1 Because of the wide variety of pH meters and accessories, detailed operating procedures cannot be incorporated into this method. Each analyst must be acquainted with the operation of each system and familiar with all instrument functions. Special attention to care of the electrodes is recommended.
 - 7.2 Each instrument/electrode system must be calibrated at a minimum of two points that bracket the expected pH of the samples and are approximately three pH units or more apart.
 - 7.2.1 Various instrument designs may involve use of a “balance” or “standardize” dial and/or a slope adjustment as outlined in the manufacturer’s instructions. Repeat adjustments on successive portions of the two buffer solutions as outlined in procedure 8.2 until readings are within 0.05 pH units of the buffer solution value.
- 8. Procedure
 - 8.1 Standardize the meter and electrode system as outlined in Section 7.
 - 8.2 Place the sample or buffer solution in a clean glass beaker using a sufficient volume to cover the sensing elements of the electrodes and to give adequate clearance for the magnetic stirring bar.
 - 8.2.1 If field measurements are being made the electrodes may be immersed directly in the sample stream to an adequate depth and moved in a manner to insure sufficient sample movement across the electrode sensing element as indicated by drift free (< 0.1 pH) readings.
 - 8.3 If the sample temperature differs by more than 2°C from the buffer solution the measured pH values must be corrected. Instruments are equipped with automatic or manual

⁽¹⁾National Bureau of Standards Special Publication 260.

compensators that electronically adjust for temperature differences. Refer to manufacturer's instructions.

- 8.4 After rinsing and gently wiping the electrodes, if necessary, immerse them into the sample beaker or sample stream and stir at a constant rate to provide homogeneity and suspension of solids. Rate of stirring should minimize the air transfer rate at the air water interface of the sample. Note and record sample pH and temperature. Repeat measurement on successive volumes of sample until values differ by less than 0.1 pH units. Two or three volume changes are usually sufficient.

9. Calculation

- 9.1 pH meters read directly in pH units. Report pH to the nearest 0.1 unit and temperature to the nearest °C.

10. Precision and Accuracy

- 10.1 Forty-four analysts in twenty laboratories analyzed six synthetic water samples containing exact increments of hydrogen-hydroxyl ions, with the following results:

pH Units	Standard Deviation pH Units	Accuracy as	
		Bias, %	Bias, pH Units
3.5	0.10	-0.29	-0.01
3.5	0.11	-0.00	
7.1	0.20	+1.01	+0.07
7.2	0.18	-0.03	-0.002
8.0	0.13	-0.12	-0.01
8.0	0.12	+0.16	+0.01

(FWPCA Method Study 1, Mineral and Physical Analyses)

- 10.2 In a single laboratory (EMSL), using surface water samples at an average pH of 7.7, the standard deviation was ± 0.1 .

Bibliography

1. Standard Methods for the Examination of Water and Wastewater, 14th Edition, p 460, (1975).
2. Annual Book of ASTM Standards, Part 31, "Water", Standard D1293-65, p 178 (1976).

METHOD 7740

SELENIUM (ATOMIC ABSORPTION, FURNACE TECHNIQUE)

1.0 SCOPE AND APPLICATION

1.1 Method 7740 is an atomic absorption procedure approved for determining the concentration of selenium in wastes, mobility-procedure extracts, soils, and ground water. All samples must be subjected to an appropriate dissolution step prior to analysis.

2.0 SUMMARY OF METHOD

2.1 Prior to analysis by Method 7740, samples must be prepared in order to convert organic forms of selenium to inorganic forms, to minimize organic interferences, and to convert samples to suitable solutions for analysis. The sample-preparation procedure varies, depending on the sample matrix. Aqueous samples are subjected to the acid-digestion procedure described in this method. Sludge samples are prepared using the procedure described in Method 3050.

2.2 Following the appropriate dissolution of the sample, a representative aliquot is placed manually or by means of an automatic sampler into a graphite tube furnace. The sample aliquot is then slowly evaporated to dryness, charred (ashed), and atomized. The absorption of lamp radiation during atomization will be proportional to the selenium concentration.

2.3 The typical detection limit for this method is 2 ug/L.

3.0 INTERFERENCES

3.1 Elemental selenium and many of its compounds are volatile; therefore, samples may be subject to losses of selenium during sample preparation. Spike samples and relevant standard reference materials should be processed to determine if the chosen dissolution method is appropriate.

3.2 Likewise, caution must be employed during the selection of temperature and times for the dry and char (ash) cycles. A nickel nitrate solution must be added to all digestates prior to analysis to minimize volatilization losses during drying and ashing.

3.3 In addition to the normal interferences experienced during graphite furnace analysis, selenium analysis can suffer from severe nonspecific absorption and light scattering caused by matrix components during atomization. Selenium analysis is particularly susceptible to these problems because of its low analytical wavelength (196.0 nm). Simultaneous background correction is required to avoid erroneously high results. High iron levels can give overcorrection with deuterium background. Zeeman background correction can be useful in this situation.

3.4 If the analyte is not completely volatilized and removed from the furnace during atomization, memory effects will occur. If this situation is detected, the tube should be cleaned by operating the furnace at full power at regular intervals in the analytical scheme.

3.5 Selenium analysis suffers interference from chlorides (>800 mg/L) and sulfate (>200 mg/L). The addition of nickel nitrate such that the final concentration is 1% nickel will lessen this interference.

4.0 APPARATUS AND MATERIALS

4.1 250-mL Griffin beaker.

4.2 10-mL volumetric flasks.

4.3 Atomic absorption spectrophotometer: Single- or dual-channel, single- or double-beam instrument with a grating monochromator, photomultiplier detector, adjustable slits, a wavelength range of 190-800 nm, and provisions for simultaneous background correction and interfacing with a strip-chart recorder.

4.4 Selenium hollow cathode lamp, or electrodeless discharge lamp (EDL): EDLs provide better sensitivity for the analysis of Se.

4.5 Graphite furnace: Any graphite furnace device with the appropriate temperature and timing controls.

4.6 Strip-chart recorder: A recorder is strongly recommended for furnace work so that there will be a permanent record and so that any problems with the analysis, such as drift, incomplete atomization, losses during charring, changes in sensitivity, etc., can easily be recognized.

4.7 Pipets: Microliter with disposable tips. Sizes can range from 5 to 1,000 μ L, as required.

5.0 REAGENTS

5.1 ASTM Type II water (ASTM D1193): Water should be monitored for impurities.

5.2 Concentrated nitric acid (HNO_3): Acid should be analyzed to determine levels of impurities. If a method blank made with the acid is <MDL, the acid can be used.

5.3 Hydrogen peroxide (30%): Oxidant should be analyzed to determine levels of impurities. If a method blank made with the oxidant is <MDL, the oxidant can be used.

5.4 Selenium standard stock solution (1.000 mg/L): Either procure a certified aqueous standard from a supplier and verify by comparison with a second standard, or dissolve 0.3453 g of selenious acid (actual assay 94.6% H_2SeO_3 , analytical reagent grade) or equivalent in Type II water and dilute to 200 mL.

5.5 Nickel nitrate solution (5%): Dissolve 24.780 g of ACS reagent grade $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ or equivalent in Type II water and dilute to 100 mL.

5.6 Nickel nitrate solution (1%): Dilute 20 mL of the 5% nickel nitrate to 100 mL with Type II water.

5.7 Selenium working standards: Prepare dilutions of the stock solution to be used as calibration standards at the time of the analysis. Withdraw appropriate aliquots of the stock solution, add 1 mL of concentrated HNO_3 , 2 mL of 30% H_2O_2 , and 2 mL of the 5% nickel nitrate solution. Dilute to 100 mL with Type II water.

5.8 Air: Cleaned and dried through a suitable filter to remove oil, water, and other foreign substances. The source may be a compressor or a cylinder of industrial-grade compressed air.

5.9 Hydrogen: Suitable for instrumental analysis.

6.0 SAMPLE COLLECTION, PRESERVATION, AND HANDLING

6.1 All samples must have been collected using a sampling plan that addresses the considerations discussed in Chapter Nine of this manual.

6.2 All sample containers must be prewashed with detergents, acids, and Type II water. Plastic and glass containers are both suitable.

6.3 Special containers (e.g., containers used for volatile organic analysis) may have to be used if very volatile selenium compounds are to be analyzed.

6.4 Aqueous samples must be acidified to a pH of <2 with nitric acid.

6.5 Nonaqueous samples shall be refrigerated, when possible, and analyzed as soon as possible.

7.0 PROCEDURE

7.1 Sample preparation: Aqueous samples should be prepared in the manner described in Steps 7.1.1 to 7.1.3. Sludge-type samples should be prepared according to Method 3050. The applicability of a sample-preparation technique to a new matrix type must be demonstrated by analyzing spiked samples and/or relevant standard reference materials.

7.1.1 Transfer 100 mL of well-mixed sample to a 250-mL Griffin beaker; add 2 mL of 30% H_2O_2 and sufficient concentrated HNO_3 to result in an acid concentration of 1% (v/v). Heat for 1 hr at 95°C or until the volume is slightly less than 50 mL.

7.1.2 Cool and bring back to 50 mL with Type II water.

7.1.3 Pipet 5 mL of this digested solution into a 10-mL volumetric flask, add 1 mL of the 1% nickel nitrate solution, and dilute to 10 mL with Type II water. The sample is now ready for injection into the furnace.

7.2 The 196.0-nm wavelength line and a background correction system must be employed. Follow the manufacturer's suggestions for all other spectrophotometer parameters.

7.3 Furnace parameters suggested by the manufacturer should be employed as guidelines. Because temperature-sensing mechanisms and temperature controllers can vary between instruments or with time, the validity of the furnace parameters must be periodically confirmed by systematically altering the furnace parameters while analyzing a standard. In this manner, losses of analyte due to overly high temperature settings or losses in sensitivity due to less than optimum settings can be minimized. Similar verification of furnace parameters may be required for complex sample matrices.

7.4 Inject a measured μL -aliquot of sample into the furnace and atomize. If the concentration found is greater than the highest standard, the sample should be diluted in the same acid matrix and reanalyzed. The use of multiple injections can improve accuracy and help detect furnace pipetting errors.

7.5 Analyze all EP extracts, all samples analyzed as part of a delisting petition, and all samples that suffer from matrix interferences by the method of standard additions.

7.6 Run a check standard after approximately every 10 sample injections. Standards are run in part to monitor the life and performance of the graphite tube. Lack of reproducibility or significant change in the signal for the standard indicates that the tube should be replaced.

7.7 Duplicates, spiked samples, and check standards should be analyzed every 20 samples.

7.8 Calculate metal concentrations: (1) by the method of standard additions, (2) from a calibration curve, or (3) directly from the instrument's concentration read-out. All dilution or concentration factors must be taken into account.

8.0 QUALITY CONTROL

8.1 All quality control data should be maintained and available for easy reference or inspection.

8.2 Calibration curves must be composed of a minimum of a blank and three standards. A calibration curve should be made for every hour of continuous sample analysis.

8.3 Dilute samples if they are more concentrated than the highest standard or if they fall on the plateau of a calibration curve.

8.4 Employ a minimum of one blank per sample batch to determine if contamination or any memory effects are occurring.

8.5 Verify calibration with an independently prepared check standard every 15 samples.

8.6 Run one spike duplicate sample for every 10 samples. A duplicate sample is a sample brought through the entire sample preparation and analytical process.

8.7 The method of standard additions (see Method 7000, Section 8.7) shall be used for the analysis of all EP extracts, on all analyses submitted as part of a delisting petition, and whenever a new sample matrix is being analyzed.

9.0 METHOD PERFORMANCE

9.1 Precision and accuracy data are available in Method 270.2 of Methods for Chemical Analysis of Water and Wastes.

9.2 The data shown in Table 1 were obtained from records of state and contractor laboratories. The data are intended to show the precision of the combined sample preparation and analysis method.

10.0 REFERENCES

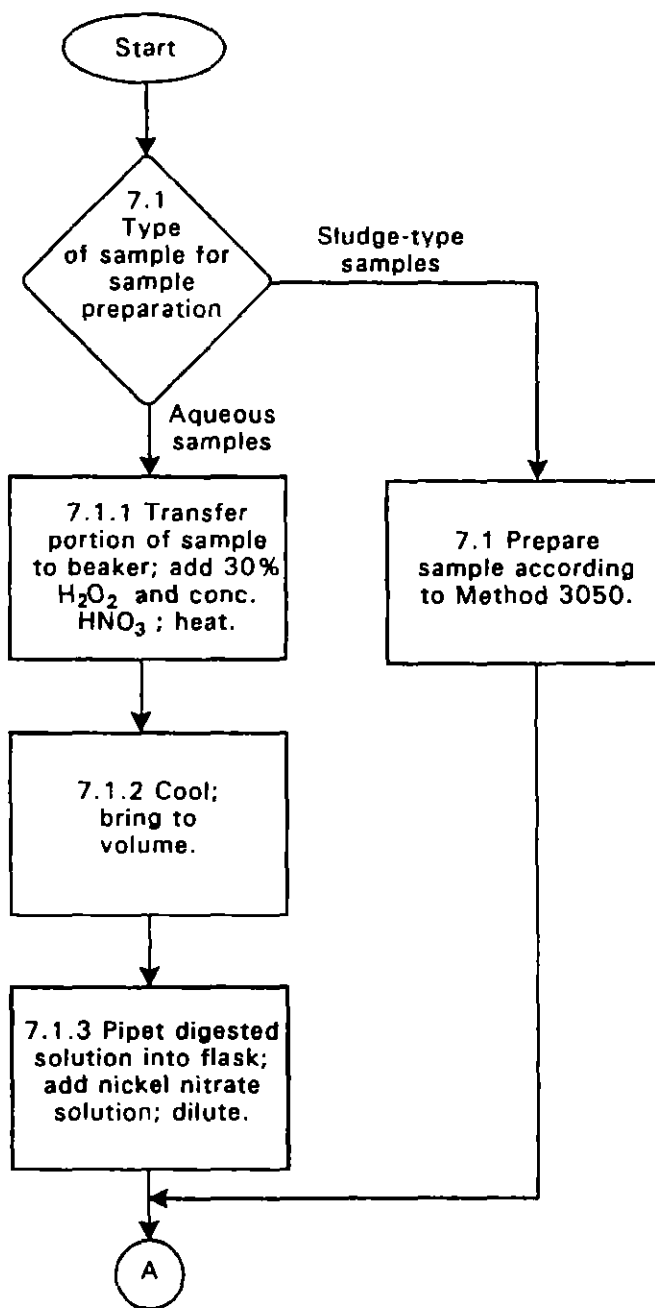
1. Methods for Chemical Analysis of Water and Wastes, EPA-600/4-82-055, December 1982, Method 270.2.

2. Gaskill, A., Compilation and Evaluation of RCRA Method Performance Data, Work Assignment No. 2, EPA Contract No. 68-01-7075, September 1986.

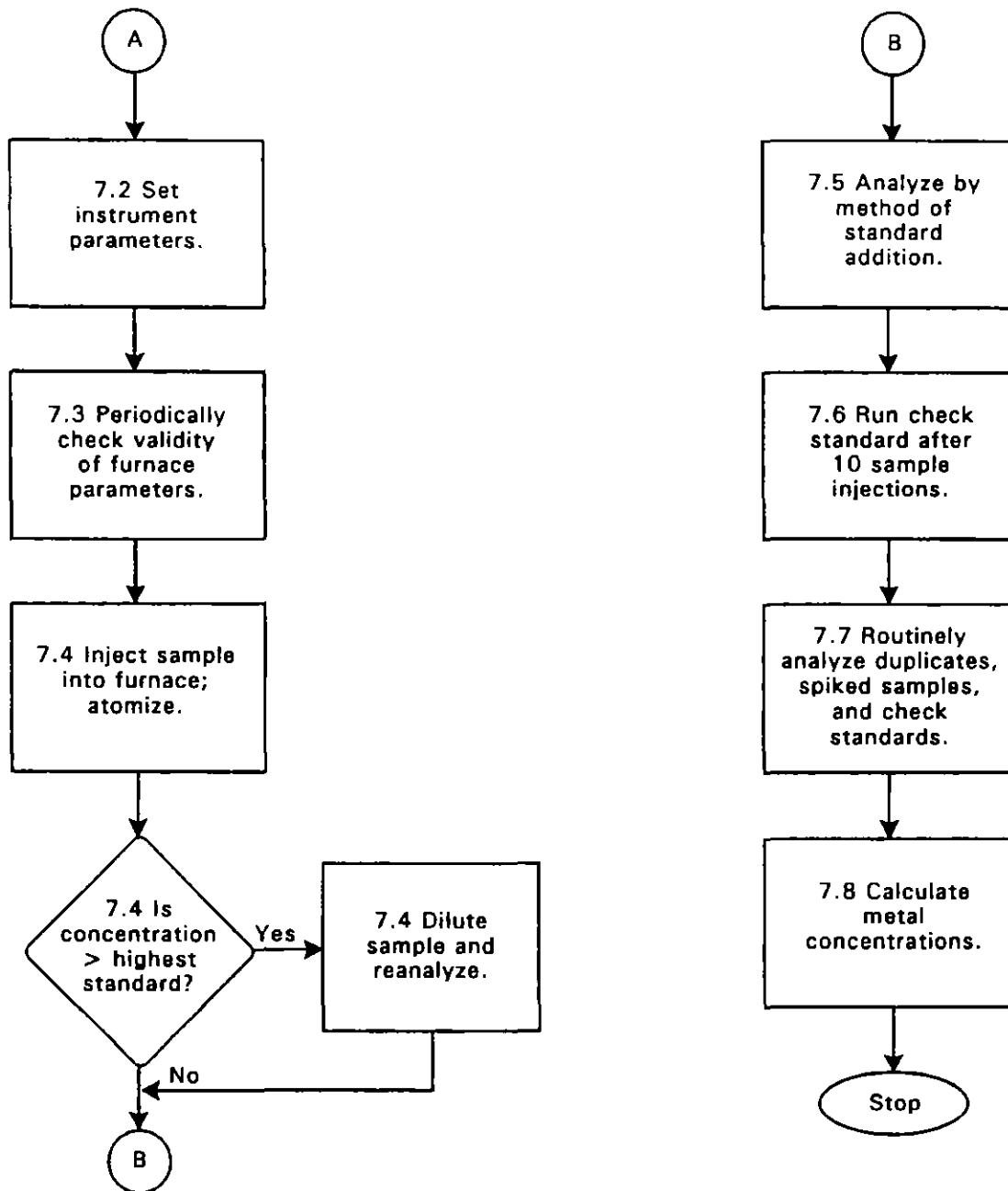
TABLE 1. METHOD PERFORMANCE DATA

Sample Matrix	Preparation Method	Laboratory Replicates
Emission control dust	3050	14, 11 ug/g

METHOD 7740
SELENIUM (ATOMIC ABSORPTION, FURNACE METHOD)



METHOD 7740
SELENIUM (ATOMIC ABSORPTION, FURNACE METHOD)
(Continued)



METHOD 7841

THALLIUM (ATOMIC ABSORPTION, FURNACE TECHNIQUE)

1.0 SCOPE AND APPLICATION

1.1 See Section 1.0 of Method 7000.

2.0 SUMMARY OF METHOD

2.1 See Section 2.0 of Method 7000.

3.0 INTERFERENCES

3.1 See Section 3.0 of Method 7000 if interferences are suspected.

3.2 Background correction is required.

3.3 Hydrochloric acid or excessive chloride will cause volatilization of thallium at low temperatures. Verification that losses are not occurring, by spiked samples or standard additions, must be made for each sample matrix.

3.4 Palladium is a suitable matrix modifier for thallium analysis.

4.0 APPARATUS AND MATERIALS

4.1 For basic apparatus, see Section 4.0 of Method 7000.

4.2 Instrument parameters (general):

4.2.1 Drying time and temp: 30 sec at 125°C.

4.2.2 Ashing time and temp: 30 sec at 400°C.

4.2.3 Atomizing time and temp: 10 sec at 2400°C.

4.2.4 Purge gas: Argon or nitrogen.

4.2.5 Wavelength: 276.8 nm.

4.2.6 Background correction: Required.

4.2.7 Other operating parameters should be set as specified by the particular instrument manufacturer.

NOTE: The above concentration values and instrument conditions are for a Perkin-Elmer HGA-2100, based on the use of a 20- μ L injection, continuous-flow purge gas, and nonpyrolytic graphite. Smaller sizes of furnace devices or those employing faster rates of atomization can be operated using lower atomization temperatures for shorter time periods than the above-recommended settings.

5.0 REAGENTS

5.1 See Section 5.0 of Method 7000.

5.2 Preparation of standards:

5.2.1 Stock solution: Dissolve 1.303 g thallium nitrate, $TlNO_3$ (analytical reagent grade), in Type II water, acidify with 10 mL concentrated HNO_3 , and dilute to 1 liter with Type II water. Alternatively, procure a certified standard from a supplier and verify by comparison with a second standard.

5.2.2 Prepare dilutions of the stock solution to be used as calibration standards at the time of analysis. The calibration standards should be prepared using the same type of acid and at the same concentrations as in the sample after processing (0.5% v/v HNO_3).

5.3 Palladium chloride: Weigh 0.25 g of $PdCl_2$ to the nearest 0.0001 g. Dissolve in 10 mL of 1:1 HNO_3 and dilute to 1 liter with Type II water. Use equal volumes of sample and palladium solution.

6.0 SAMPLE COLLECTION, PRESERVATION, AND HANDLING

6.1 See Chapter Three, Section 3.1.3, Sample Handling and Preservation.

7.0 PROCEDURE

7.1 Sample preparation: The procedures for preparation of the sample are given in Chapter Three, Section 3.2.

7.2 See Method 7000, Paragraph 7.3, Furnace Procedure. The calculation is given in Method 7000, Paragraph 7.4.

8.0 QUALITY CONTROL

8.1 See Section 8.0 of Method 7000.

9.0 METHOD PERFORMANCE

9.1 Precision and accuracy data are not available at this time.

9.2 The performance characteristics for an aqueous sample free of interferences are:

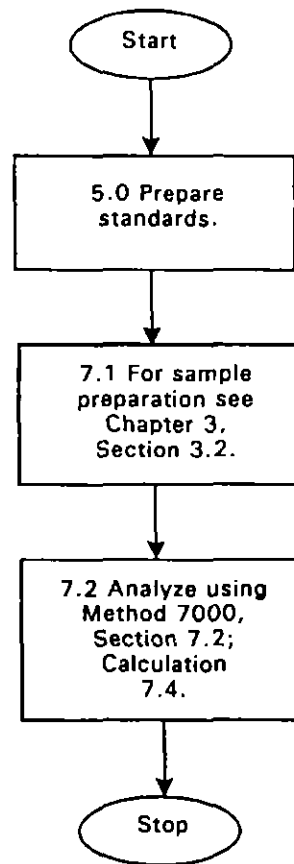
Optimum concentration range: 5-100 ug/L.

Detection limit: 1 ug/L.

10.0 REFERENCES

1. Application of Matrix-Modification in Determination of Thallium in Wastewater by Graphite-Furnace Atomic-Absorption Spectrometry, Talanta, 31(2) (1984), pp. 150-152.

METHOD 7841
THALLIUM (ATOMIC ABSORPTION, FURNACE TECHNIQUE)



Attachment B

Surface Water Sampling Procedure

B-1. Surface Water Sampling Procedure

B-1.1 Purpose

This procedure presents a protocol for surface-water sampling. It provides a standard sampling methodology to assure uniformity and quality of surface-water samples. This procedure outlines the responsibilities for field personnel involved in surface-water sampling activities.

B-1.2 Equipment and Materials

Surface water will be sampled using either pre-cleaned, disposable samplers or decontaminated sampling equipment. Sampling equipment must be compatible with the surface-water to be sampled and for the parameters to be analyzed. In general, stainless steel equipment can be used for sampling both metals and organic constituents; plastic equipment can be used for sampling metals. Either type of sampling device can be used for this project.

Equipment and materials for surface-water sampling include the following:

- Sample collection equipment (i.e. cup or other open-ended container),
- Decontamination materials (i.e. potable water & phosphate-free soap),
- On-site measurement instruments (i.e. temperature, pH, conductivity meter), and
- Laboratory provided sample containers, coolers, and blue ice.

B-1.3 Equipment Decontamination

If non-disposable sampling equipment is used to obtain the surface water samples, the following equipment decontamination procedure will be used to decontaminate the sampling equipment prior to the collection of each sample.

B-1.3.1 Equipment and Materials

The equipment needed for equipment decontamination includes:

- distilled or potable water,
- phosphate-free detergent,
- brushes, and
- latex, or similar, gloves.

B-1.4 Procedure

B-1.4.1 Sampling Equipment

Portions of sampling and test equipment which contact the sample are thoroughly cleaned before use for sampling. Equipment must be cleaned before and after a sample is taken. This includes water-level tapes or probes, pumps, tubing, bailers, lifting line, field analytical instruments for on-site use, and other equipment or portions thereof that are to be immersed or contacted. The procedures for equipment cleaning are as follows, unless otherwise specified by the equipment manufacturer:

- brush to remove any residue,
- clean with a solution of distilled or potable water and phosphate-free laboratory detergent,
- rinse with distilled water, and
- air dry.
- use equipment immediately, or pack equipment in plastic wrap or bags, or in aluminum foil.

Equipment cleaned prior to field use is cleaned again after transfer to the sampling facility.

Pre-cleaned sample containers will be provided by the analytical laboratory.

Any necessary deviation from these procedures is documented in the permanent record of the sampling episode.

B-1.5 Surface-Water Sampling Procedure

Surface-water sampling will be conducted to determine the surface water quality at the established monitoring and compliance points. A summary of the surface water sampling procedure is as follows:

- pre-arrangement with testing laboratory;
- assembly and preparation of sampling equipment and supplies;
- surface-water sampling:
 - visual inspection of water,
 - on-site measurement of parameters, and
 - sampling;
- sample packaging and shipment:
 - sample labeling, and
 - sample packaging;
- completion of sample records;

- completion of Chain-of-Custody records; and
- sample shipment.

B-1.6 Sampling Personnel

Sampling will be conducted by personnel trained in the this surface water sampling procedure. Sampling personnel will have thoroughly reviewed this procedure prior to each sampling event.

B-1.7 Arrangements with Analytical Laboratory

Prior to sampling, arrangements will be made with the analytical laboratory that will conduct the sample analyses to make the necessary arrangements for the analyses and to obtain the appropriate sample containers, preservatives, and shipping containers.

The laboratory will provide a sufficient number of sample containers for the surface-water locations to be sampled and the QC samples to be included. The laboratory will determine the proper type and size for the containers, based upon the analyses to be conducted. For samples requiring chemical preservation, preservatives will be added to containers by the laboratory prior to shipping containers to the facility. Shipping containers, ice chests with adequate container padding and cooling media, will be sent to the facility with the containers.

B-1.8 Preparation for Sampling

Equipment will be controlled, cleaned, and maintained in accordance equipment decontamination procedures specified in Section 1.3 of this procedure.

B-1.9 Sample Collection

The surface water samples will be collected in a manner to prevent contamination of the surface-water being sampled and the samples themselves. The primary way such contamination can occur is through contact with improperly cleaned equipment. To prevent such contamination, sampling equipment and sample containers will be thoroughly cleaned before and after field use and between uses at different sampling locations as described in the equipment decontamination procedures (Section 1.3). In addition to the use of properly cleaned equipment, two further precautions will be followed.

- A clean pair of new, disposable latex (or similar) gloves will be worn at each different location.
- Field personnel will minimize the potential for cross-contamination that could be caused by trace constituents the might occur on their waders or rubber boots by using a long-handled sampling cup or bailer to collect the sample away from the area disturbed by wading.

The following paragraphs present procedures for the activities which comprise surface-water sample acquisition. These activities will be performed in the same order as presented below. Exceptions to this procedure will be noted in the permanent sampling record.

B-1.9.1 Preparation of Location

Prior to starting the sampling procedure, the area to be sampled should be cleared of foreign materials such as brush, rocks, debris, etc. A clean (new), disposable, plastic sheet should be placed at the sample location. This preparation will prevent sampling equipment from inadvertently contacting the ground.

B-1.9.2 Visual Inspection of Surface-Water

Prior to sampling, a small quantity of water will be obtained and inspected for the presence of a floating film, suspended sediment, or other indications of potential contamination. Any sample odors will be noted. These observations regarding odor or visual evidence of contamination will be recorded in the sampling record.

B-1.9.3 Sample Collection

Grab samples are taken from surface-water bodies from either the shore or by wading into the water. A bailer or long-handled cup constructed of stainless steel or Teflon can be used to collect water samples. Care will be taken during sampling to prevent undue disturbance of the water. The sampler will be lowered into and extracted from the water gently to prevent splashing. The sample may be poured directly into the appropriate containers. While pouring water from a sampler, the water will be carefully poured down the inside of the sample bottle to prevent significant aeration of the sample.

Unless other instructions are given by the analytical laboratory, the sample containers will be completely filled so that no air space remains in the container.

B-1.9.4 Quality Control Samples

Surface-water samples will include one duplicate sample for every ten locations sampled. The decision of which sample to duplicate will be made by sampling personnel. The duplicate sample will be given a designation which cannot be confused with other samples to be tested and will be analyzed for the same constituents as the regular samples.

B-1.9.5 On-Site Parameter Measurement

Certain chemical and physical parameters in water can change significantly within a short time of sample acquisition. These will be measured on-site with portable equipment. These parameters are:

- pH,
- specific conductance, and
- temperature.

These parameters will be obtained in accordance with the field measurement procedures specified in Attachment B or as directed by the equipment manufacturer. These parameters will be measured in unfiltered, unpreserved, "fresh" water taken by the same technique as the samples taken for laboratory analyses. The measurements will be made in a clean glass container separate from those intended for laboratory analysis. The tested sample will be discarded after use. The measured values will be recorded in the sampling record.

B-1.10 *Sample Handling and Chain-of-Custody*

The samples will be prepared, labeled, stored, and shipped in accordance with the procedures specified below.

B-1.10.1 Sample Containers and Sample Labeling

The samples will be placed in pre-clean sample bottles supplied by the laboratory that include the appropriate preservative (if required). The surface water samples will be placed in sample containers provided by the analytical laboratory. Containers are expected to be pre-cleaned by the laboratory and shipped sealed and ready for use. At a minimum, the sample bottles will be labeled with the sample number, sample date and time, and sampler name at each sample location.

Sample containers are labeled with a label firmly attached to the container side (not lid). The following information is to be legibly and indelibly written on the sample label attached to the sample container side:

- project or facility name (including project number),
- sample type,
- sample location number,
- sampling date and time, and
- sample collector's name or initials.

B-1.10.2 Sample Preservation

After collection, the samples will be placed in a cooler with blue ice and maintained at approximately 4° Centigrade (C) until delivered to the analytical laboratory.

B-1.10.3 Sampling Records

To provide complete documentation of on-site aspects of the sampling episode, a detailed sampling record is to be prepared for samples. These records will include the following information:

- sample location, including facility name and sample number or well number,
- date and time of sampling,
- sampler's identification,
- weather conditions,
- sample description (including type and volume),
- sampling method,

- well evacuation method and volume of ground water evacuated (if applicable),
- depth to ground-water surface (if applicable),
- field observations and measurements of:
 - sample appearance, color, turbidity, oil film, etc.,
 - sample odor,
 - temperature,
 - pH,
 - specific conductance,
- sample preparation methods,
- shipping receipts, and
- any other information which may be pertinent.

B-1.10.4 Chain-of-Custody Documentation

After samples have been obtained, placed into containers, and labeled, Chain-of-Custody procedures will be followed to establish a written record concerning sample movement between the sampling site and the testing laboratory or sample storage location. A Chain-of-Custody is completed for each shipping container by the person who conducts the sampling and packing of the samples. Chain-of-Custody documentation for each container is completed in triplicate. One copy of this form will be maintained at the facility. The other two copies are sealed in an envelope attached to the container and shipped to the laboratory or sample storage location. One of the copies shipped to the laboratory or sample storage location is returned to the facility and becomes part of the permanent record of the sample.

A copy of a sample Chain-of-Custody form is included as Figure B-2. At a minimum, the form must contain the following information:

- project name,
- sample number,
- signature of collector,
- date and time of collection,
- sample location,
- sample type,
- preservatives added (if appropriate),
- analyses requested,

- signatures involved in chain of possession, and
- inclusive dates of possession.

B-1.10.5 Sample Shipment

Properly-completed Chain-of-Custody documentation is prepared and placed inside the shipping container in a waterproof plastic bag, so that it will be immediately apparent to the laboratory personnel receiving the container, but will not be damaged or lost during shipping. The shipping containers will be sealed, so that it will be obvious if the seal has been tampered with or broken.

A Chain-of-Custody form will be completed for each shipping container. Each time the sample bottles or samples change hands, both the sender and the receiver need to sign and date the Chain-of-Custody form.. The completed Chain-of-Custody becomes a permanent record of the sample handling and shipping.

Samples are transported by sampling personnel directly to the laboratory or are shipped via an express carrier (i.e., Federal Express). Shipping instructions and receipts provided to the sender and addressee by the shipper will become part of the permanent record. Such instructions and receipts should be returned to the facility.

B-1.11 Sampling Records

To provide complete documentation of sampling, detailed records will be maintained during sampling. These records will include the information listed below:

- sample location (facility name);
- sample identification (sample area name and sample number);
- sample location map or detailed sketch;
- date and time of sampling;
- sampling method;
- field observations of:
 - sample appearance (i.e., color, suspended sediment), and
 - sample odor;
- weather conditions;
- sampler's identification; and
- any other information which is significant.

Surface-water sampling information will be recorded on the surface-water sampling record (Figure B-1).

Figure B-1
SurfaceWater Sampling Record

FIGURE B-1
SURFACE-WATER SAMPLING RECORD

SAMPLING LOCATION: _____

SAMPLING DATE: _____

REASON FOR SAMPLING: REGULAR SAMPLING _____
 SPECIAL SAMPLING _____

TIME AND DATE OF SAMPLING: _____ am / pm _____ 20 _____

SAMPLE COLLECTED BY: _____

WEATHER: _____

DATUM FOR WATER LEVEL MEASUREMENT (describe): _____

EQUIPMENT CLEANING MATERIALS (list): _____

WATER SURFACE (ABOVE/BELOW) DATUM: _____

SAMPLING METHOD: _____

OBSERVATIONS OF WATER:

Appearance: _____

Odor: _____

Other Comments: _____

Depth: _____

ON-SITE MEASUREMENTS:

Temperature _____ °F °C Measured with _____

pH _____ Measured with _____

Conductivity _____ Measured with _____

Other _____

SAMPLE CONTAINERS (material, number, size) _____

ON-SITE HANDLING:

Filtration: Method _____ No. of Containers _____

Method _____ No. of Containers _____

Method _____ No. of Containers _____

Preservatives: Method _____ No. of Containers _____

Method _____ No. of Containers _____

Method _____ No. of Containers _____

Method _____ No. of Containers _____

CONTAINER HANDLING: Sides Labeled _____
 Lids Taped _____
 Samples Packed in Ice _____

OTHER COMMENTS: _____

SAMPLER'S SIGNATURE: _____ DATE: _____

Figure B-2
Example Chain-of-Custody

Example Chain-of-Custody

CHAIN OF CUSTODY RECORD

[illegible]